

VOLUME III



THE SCENT HANDBOOK

FORMULATION

*Precision weighing for small-batch EDP blending – a
bench reference for the 0.01 g scale*

TABLE OF CONTENTS

01	The Principle <i>Why mass, not volume</i>	03
02	The Calculation <i>Deriving oil and alcohol mass from target volume</i>	04
03	Quick Reference <i>Direct-weighing table – 30/50/100 ml, four concentrations</i>	05
04	Bench Setup <i>Equipment and workspace checklist</i>	06
05	Weighing the Oil <i>Tare, add, confirm</i>	07
06	Adding the Alcohol <i>Second tare, fill to target</i>	08
07	Mixing Procedure <i>Sealing and blending the batch</i>	09
08	The Five Precision Rules <i>Discipline at 0.01 g resolution</i>	10
09	Quality Control <i>Pre-release checklist</i>	11
10	Bench Workflow <i>The process on one page</i>	12
11	Worked Examples <i>Three full calculations</i>	13
12	On Accuracy <i>Limits of the reference densities, and a closing note</i>	14

THE PRINCIPLE

Why professional formulation is done by weight, not by volume.

Professional fragrance formulation is carried out by mass (grams), not volume (millilitres). Volume shifts slightly with temperature — a measured 25 ml at a warm room is not quite the same 25 ml in a cool one. Mass does not move. A precision scale gives a repeatable measurement that a graduated cylinder cannot.

This handbook works from two reference densities. They are calculation assumptions for bench use — not universal constants.

REFERENCE DENSITIES USED IN THIS HANDBOOK

MATERIAL	REFERENCE DENSITY
Fragrance oil / concentrate	0.95 g/ml
96% ethanol	0.81 g/ml

IMPORTANT

Actual fragrance concentrates vary in density by material and supplier. For production-grade accuracy, use the supplier's measured density, or determine the density of the specific batch you are working with. Treat the table in this handbook as a bench-ready starting point, not a certificate of analysis.

THE CALCULATION

From a target volume and concentration to two numbers on the scale.

For a target volume V and oil concentration C , the component volumes are found first, then converted to mass using each material's reference density.

$$\text{Oil volume} = V \times C$$

$$\text{Alcohol volume} = V \times (1 - C)$$

$$\text{Oil mass} = V \times C \times 0.95$$

$$\text{Alcohol mass} = V \times (1 - C) \times 0.81$$

The final batch weight is simply the sum of the two component masses.

WORKED — 100 ML AT 25%

100 ML • 25% EDP INTENSE

reference example

23.75 g

OIL (25 ML $\times$ 0.95)

60.75 g

ALCOHOL (75 ML $\times$ 0.81)

84.50 g

TOTAL BATCH WEIGHT

Section 03 carries this calculation through for the bottle sizes and concentrations used most often at the bench, so it rarely needs to be worked out by hand.

QUICK REFERENCE

Direct-weighing values for the four bottle sizes and concentrations used most.

BOTTLE	CONCENTRATION	OIL	ALCOHOL	TOTAL
30 ml	15% EDT	4.28 g	20.66 g	24.94 g
	20% EDP	5.70 g	19.44 g	25.14 g
	25% EDP Intense	7.13 g	18.23 g	25.36 g
	30% Extrait	8.55 g	17.01 g	25.56 g
50 ml	15% EDT	7.13 g	34.43 g	41.56 g
	20% EDP	9.50 g	32.40 g	41.90 g
	25% EDP Intense	11.88 g	30.38 g	42.26 g
	30% Extrait	14.25 g	28.35 g	42.60 g
100 ml	15% EDT	14.25 g	68.85 g	83.10 g
	20% EDP	19.00 g	64.80 g	83.80 g
	25% EDP Intense	23.75 g	60.75 g	84.50 g
	30% Extrait	28.50 g	56.70 g	85.20 g

READING THIS TABLE

Figures are calculated at 0.95 g/ml (oil) and 0.81 g/ml (96% ethanol) — see Section 01. This page is reproduced as a standalone bench card at the back of this set for use next to the scale.

BENCH SETUP

What to have in place before the scale is switched on.

EQUIPMENT

- | | |
|--|--|
| ☐ Precision digital scale, 0.01 g readability | ☐ Pipette, syringe, or suitable transfer tool |
| ☐ Clean fragrance bottle or mixing vessel | ☐ Paper towels / lint-free wipes |
| ☐ Fragrance concentrate / oil | ☐ Labels and permanent marker |
| ☐ 96% ethanol | ☐ Gloves and eye protection |

WORK AREA CHECKLIST

Clean → Dry → Stable → Ventilated → Away from ignition sources

CAUTION — FLAMMABLE

Ethanol is highly flammable. Keep the work area away from flames, sparks, cigarettes, hot surfaces, and other ignition sources. Follow the safety information supplied with your ethanol and fragrance materials.

WEIGHING THE OIL

The first of two components – tare, add, confirm.

1

PREPARE THE BOTTLE

Make sure the bottle and equipment are clean and completely dry. Place the empty bottle or mixing vessel centrally on the scale.

2

TARE THE SCALE

Wait for the reading to stabilize, then press TARE. The display should return to 0.00 g.

3

ADD THE OIL

Slowly add the fragrance concentrate until the scale reaches the exact oil target from Section 03. Add slowly near the target – do not correct a large overshoot by removing material unless you have a controlled procedure for it.

Example – 100 ml / 25% → 23.75 g oil

4

CONFIRM

Pause and confirm the reading is stable. Record the actual weight if you are keeping a batch record.

ADDING THE ALCOHOL

The second component – a fresh tare, then fill to target.

5

TARE AGAIN

With the oil already in the vessel, press TARE again. The scale should return to 0.00 g while retaining the oil already in the vessel.

6

ADD ETHANOL

Slowly add the required ethanol weight from Section 03, stopping at the target and allowing the display to stabilize.

Example – 100 ml / 25% → 60.75 g ethanol

7

CLOSE IMMEDIATELY

Seal the bottle or vessel securely to minimize evaporation and contamination.

RULE 1 – ALWAYS TARE FRESH

Never mentally subtract the previous component's weight. Tare the scale and weigh the next component directly. See Section 08 for the full set of precision rules.

MIXING PROCEDURE

Sealing and blending the finished batch.

- ☐ Confirm the container is sealed.
- ☐ Hold the bottle securely.
- ☐ Mix with a gentle, controlled motion for approximately 60 seconds.
- ☐ Avoid violent shaking that can introduce unnecessary air bubbles.
- ☐ Wipe the exterior clean and label the batch.

SUGGESTED LABEL

Fragrance	
Concentration	
Bottle size	
Oil lot / batch	
Date	
Oil weight	----- g
Alcohol weight	----- g
Total weight	----- g

A FULL-SIZE PRINTABLE LABEL SHEET IS INCLUDED AS A STANDALONE CARD IN THIS SET.

THE FIVE PRECISION RULES

Discipline at 0.01 g resolution.

RULE 1

TARE BEFORE EVERY COMPONENT

Never mentally subtract the previous component. Tare the scale and weigh the next component directly.

RULE 2

ADD SLOWLY

At 0.01 g resolution, the last few tenths of a gram require controlled, deliberate dispensing.

RULE 3

STABILIZE BEFORE RECORDING

Air movement, vibration, warm containers, and touching the vessel can all make the reading fluctuate.

RULE 4

KEEP COMPONENTS AT A CONSISTENT TEMPERATURE

The calculations assume the stated reference densities. Temperature affects density, and therefore the relationship between volume and mass.

RULE 5

RECORD ACTUAL WEIGHTS

For repeatability, record what was actually weighed — not only the intended target.

QUALITY CONTROL

Verify before releasing a finished batch.

- ☐ Correct bottle size selected
- ☐ Correct concentration selected
- ☐ Correct oil weight used
- ☐ Correct alcohol weight used
- ☐ Scale was tared correctly
- ☐ Components were clean and uncontaminated
- ☐ Container was sealed after mixing
- ☐ Batch was labelled
- ☐ Date and batch details were recorded
- ☐ Appearance checked for unexpected cloudiness, separation, or particles

IF SOMETHING LOOKS UNUSUAL

Do not assume it is acceptable. Check the raw-material specifications and your formulation procedure before use.

THIS CHECKLIST IS REPRODUCED AS A STANDALONE BENCH CARD IN THIS SET.

BENCH WORKFLOW

The full process, on one page.

PREPARE

01

Clean equipment, prepare materials, ventilate the workspace, keep ethanol away from ignition sources.

WEIGH OIL

02

Bottle on scale → TARE → add oil → stabilize → record.

WEIGH ALCOHOL

03

TARE again → add ethanol → stabilize → record.

MIX

04

Seal the vessel → mix gently for approximately 60 seconds.

FINISH

05

Label → record batch → inspect → store appropriately.

WORKED EXAMPLES

Three complete calculations, cross-referenced against Section 03.

30 ML · 20% EDP

worked example

5.70 g	19.44 g	25.14 g
OIL	ALCOHOL	TOTAL

50 ML · 25% EDP INTENSE

worked example

11.88 g	30.38 g	42.26 g
OIL	ALCOHOL	TOTAL

100 ML · 30% EXTRAIT

worked example

28.50 g	56.70 g	85.20 g
OIL	ALCOHOL	TOTAL

ON ACCURACY

The limits of the reference densities used throughout this handbook.

This handbook is a practical weighing reference built on the stated density assumptions of 0.95 g/ml for fragrance oil and 0.81 g/ml for 96% ethanol.

For professional or commercial production, do not treat the quick-reference table as universally applicable to every concentrate. Different concentrates carry different densities, and the actual final volume after blending can differ from the nominal sum of component volumes. Validate specific materials and the formulation method whenever accuracy is critical.

“Precision is not about weighing more carefully once. It is about using the same controlled method every time.”